NON-PARAMETRIC ANALYSIS OF HEART STROKE DATASET

Presented by

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INTRODUCTION

- This project explores a stroke prediction dataset using non-parametric statistical methods.
- Unlike traditional techniques, these methods don’t assume that the data follows a normal pattern, which makes them ideal for medical datasets that often behave unpredictably.
- We focus on understanding how factors like age, BMI, glucose level, hypertension, heart disease, and lifestyle habits relate to stroke risk.
- By using flexible, assumption-free tests, our goal is to uncover meaningful patterns that can help identify key factors influencing stroke occurrence.

DATASET

- The dataset used is the Healthcare Stroke Dataset, publicly available on Kaggle
- Clinical and demographic details of 5,110 individuals with 12 attributes describing personal, lifestyle, and medical factors provide a comprehensive view of patient characteristics.

Feature	Type	Description
id	Identifier	Unique ID for each individual
gender	Categorical	Male, Female, or Other
age	Continuous	Patient’s age in years (0–82)
hypertension	Binary (0/1)	1 = Yes, 0 = No
heart_disease	Binary (0/1)	1 = Yes, 0 = No
ever_married	Categorical	Marital status (Yes/No)
work_type	Categorical	Private, Self-employed, Govt job, Children, Never worked
residence_type	Categorical	Urban or Rural
avg_glucose_level	Continuous	Average blood glucose (mg/dL)
bmi	Continuous	Body Mass Index (with 201 missing values)
smoking_status	Categorical	Formerly smoked, never smoked, smokes, or unknown
stroke	Binary (0/1)	1 = Stroke occurred, 0 = No stroke

OBJECTIVE

- To identify and interpret meaningful patterns within the clinical and demographic variables associated with stroke occurrence.
- To apply the **Run Test** to assess the randomness in key continuous variables such as age, BMI, and average glucose level.
- To use the **Kolmogorov–Smirnov (K–S) Test** to examine whether continuous variables follow a normal distribution.
- To perform the **Mann–Whitney U Test** to detect significant differences in BMI between stroke and non-stroke patient groups.
- To conduct correlation analysis (**Spearman Rank and Kendall's Tau Tests**) to explore relationships among continuous variables such as age, BMI, and average glucose level.
- To apply the **Chi-Square Test** to examine associations between categorical variables (e.g., gender, hypertension, heart disease, smoking status, residence type) and stroke occurrence.
- To use the **Kruskal–Wallis Test** to determine whether the median BMI or glucose level differs significantly across groups such as work type or smoking status.

METHODS

- **Run Test:** Used to check the randomness of data sequences and identify if variables show any specific pattern or dependency
- **Kolmogorov–Smirnov (K–S) Test:** Compares the sample distribution with a reference (e.g., normal) distribution to check whether data follows that pattern
- **Mann–Whitney U Test:** A non-parametric alternative to the t-test, used to detect significant differences between two independent groups (e.g., stroke vs. non-stroke)
- **Spearman Rank Correlation Test:** Measures the strength and direction of association between two continuous or ordinal variables without assuming normality
- **Chi-Square Test for Independence:** Examines the relationship between categorical variables, such as gender, smoking status, or hypertension with stroke occurrence
- **Kruskal–Wallis Test:** An extension of the Mann–Whitney test used to determine if three or more groups differ significantly in their median values.

1. EXAMINATION OF RANDOMNESS OF GIVEN VARIABLES USING RUNS TEST

to check the randomness of Age, BMI, and Average Glucose Level.

Age

Standardized Runs Statistic	-4.268192
P-Value	1.97e-05

BMI

Standardized Runs Statistic	0.279809
P-Value	0.7796

Avg. glucose

Standardized Runs Statistic	-0.778689
P-Value	0.436

Interpretation: BMI and Average Glucose Level show random patterns, while Age is non-random, indicating a systematic trend or dependency in its values.

2. EXAMINATION OF ASYMPTOTIC RANDOMNESS USING THE RUNS TEST

Applied to examine the randomness of the residuals. The hypothesis :

- H_0: Residuals are random (no pattern)
- H_a: Residuals are not random (show a pattern)

Residuals (Asymptotic Randomness Test)

Z-Statistic	-10.1098
P-Value	4.9963799758121539e-24

Interpretation: The p-value (4.996 × 10−24) is significantly less than 0.05, we reject the null hypothesis (H0) and conclude that the residuals exhibit non-randomness, suggesting possible autocorrelation or model misspecification..

3. ONE-SAMPLE KOLMOGOROV-SMIRNOV TEST RESULTS

To check if continuous variables follow a normal distribution. The H_0: Data is normally distributed and $\alpha = 0.05$ (significance level).

Age

KS Statistic	0.0507
P-Value	7.203626e-12

BMI

KS Statistic	0.0588
P-Value	3.444187e-15

Avg. glucose

KS Statistic	0.1828
P-Value	5.683370e-150

Interpretation: All three variables have p-values that are extremely small (e.g., 7.203626e-12), we reject null hypothesis and conclude three variable are not normally distributed

4. MANN–WHITNEY U TEST FOR THE COMPARISON OF BMI BETWEEN STROKE AND NON-STROKE PATIENTS

- H0 : The distributions of BMI for stroke and non-stroke patients are identical.
- Ha : The distributions of BMI for stroke and non-stroke patients are not identical

One-Tailed Test (Alternative: Ha: BMIstroke > BMInon-stroke)

Test Statistic (U)	694474.500
P-Value	0.00004
Conclusion	Reject $H_0 \Rightarrow$ Stroke patients have significantly higher BMI.

Interpretation: p-value is less, indicating that stroke patients tend to have higher BMI values compared to non-stroke patients.

One-Tailed Test (Alternative: Ha: BMIstroke < BMInon-stroke)

Test Statistic (U)	694474.500
P-Value	0.99996
Conclusion	Fail to Reject $H_0 \Rightarrow$ Stroke patients do not have significantly lower BMI.

Interpretation: p-value of 0.99996, confirming that stroke patients do not have lower BMI than non-stroke patients.

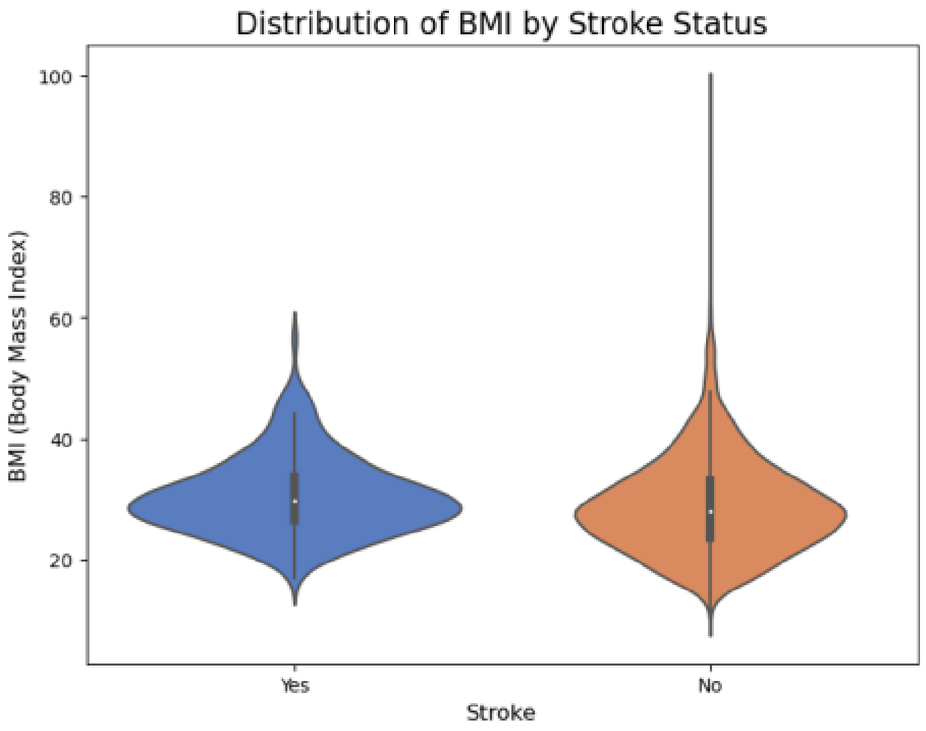
Two-Tailed Test

Test Statistic (U)	694474.500
P-Value	0.00008
Conclusion	Reject $H_0 \Rightarrow$ Significant difference in BMI between stroke and non-stroke groups.

Interpretation: p-value is less than the α = 0.05, we reject the null hypothesis (H0) and conclude that there is a statistically significant difference in BMI between stroke and non-stroke groups.

4. MANN–WHITNEY U TEST FOR THE COMPARISON OF BMI BETWEEN STROKE AND NON-STROKE PATIENTS

- The graph shows that higher BMI is linked to a greater chance of stroke.
- Stroke patients generally have higher BMI values.
- Non-stroke individuals tend to have lower BMI.
- The BMI distribution for non-stroke patients is stochastically smaller than that of stroke patients.



Distribution of BMI among Stroke and Non-Stroke Patients

5. SPEARMAN'S RANK CORRELATION AND KENDALL'S TAU TESTS TO CHECK DEPENDENCIES BETWEEN CONTINUOUS VARIABLES

- H0: There is no correlation b/w variables
- Ha: There is a significant correlation between variables.

Age vs Average Glucose Level

Spearman Rank Correlation Coefficient (ρ)	0.145168
P-Value (Spearman)	1.80×10^{-25}
Kendall's Tau (τ)	0.097858
P-Value (Kendall)	1.91×10^{-25}

Interpretation: weak but statistically significant positive correlation. Thus, as age increases, the average glucose level tends to rise slightly.

Age vs BMI

Spearman Rank Correlation Coefficient (ρ)	0.363661
P-Value (Spearman)	1.40×10^{-159}
Kendall's Tau (τ)	0.246469
P-Value (Kendall)	3.43×10^{-151}

Interpretation: moderate positive and statistically significant correlation. Implies that BMI tends to increase as age increases.

BMI vs Average Glucose Level

Spearman Rank Correlation Coefficient (ρ)	0.113696
P-Value (Spearman)	3.58×10^{-16}
Kendall's Tau (τ)	0.076055
P-Value (Kendall)	4.25×10^{-16}

Interpretation: weak but statistically significant positive relationship. Implies higher BMI tend to have slightly higher glucose levels.

6. CHI-SQUARE TEST OF INDEPENDENCE BETWEEN CATEGORICAL VARIABLES

- H0 : There is no association between variables
- Ha : There is a significant association between variables

Gender and Stroke

Chi-Square (χ^2)	0.4726
Degree of Freedom	2
P-Value	0.7895

Interpretation: weak but statistically significant positive correlation. Thus, as age increases, the average glucose level tends to rise slightly.

Hypertension and Stroke

Chi-Square (χ^2)	81.6054
Degree of Freedom	1
P-Value	1.66×10^{-19}

Interpretation: highly significant association between hypertension and stroke. Individuals with hypertension are significantly more likely to experience a stroke.

Heart Disease and Stroke

Chi-Square (χ^2)	90.2596
Degree of Freedom	1
P-Value	2.09×10^{-21}

Interpretation: weak but statistically significant positive relationship. Implies higher BMI tend to have slightly higher glucose levels.

6. CHI-SQUARE TEST OF INDEPENDENCE BETWEEN CATEGORICAL VARIABLES

- H0 : There is no association between variables
- Ha : There is a significant association between variables

Residence Type and Stroke

Chi-Square (χ^2)	1.0816
Degree of Freedom	1
P-Value	0.2983

Interpretation: no significant association between residence type (urban or rural) and stroke occurrence.

Smoking Status and Stroke

Chi-Square (χ^2)	29.1473
Degree of Freedom	3
P-Value	2.09×10^{-6}

Interpretation: significant relationship between smoking status and stroke. This suggests that smoking habits are associated with stroke occurrence.

7. KRUSKAL–WALLIS H-TEST TO DETERMINE THE DIFFERENCE BETWEEN THE MEDIANS OF TWO OR MORE INDEPENDENT GROUPS

- H0 : The median BMI is the same across all work type or smoking status groups.
- Ha : At least one work type or smoking status group has a different median BMI.

BMI vs Work Type

K-W Statistic	83.4909
Degree of Freedom	4
P-Value	3.17×10^{-17}

Interpretation: reject the null hypothesis and can say that significant difference in the median BMI among different work type groups.

BMI vs Smoking Status

K-W Statistic	13.5387
Degree of Freedom	3
P-Value	1.15×10^{-3}

Interpretation: reject the null hypothesis and can say that there is a significant difference in the median BMI across smoking status categories.

7. KRUSKAL–WALLIS H-TEST TO DETERMINE THE DIFFERENCE BETWEEN THE MEDIANS OF TWO OR MORE INDEPENDENT GROUPS

- H0 : The median average glucose level is the same across all work type or smoking status groups.
- Ha : At least one smoking status group has a different median average work type or glucose level.

Average Glucose Level vs Work Type

K-W Statistic	3.7285
Degree of Freedom	4
P-Value	0.4439

Interpretation: fail to reject the null hypothesis, indicating no significant difference in the median average glucose level among different work type groups.

Average Glucose Level vs Smoking Status

K-W Statistic	2.1909
Degree of Freedom	3
P-Value	0.3344

Interpretation: fail to reject the null hypothesis, indicating no significant difference in the median average glucose levels among smoking status groups.

CONCLUSION

- All the combined results demonstrate that non-parametric methods are appropriate for this dataset.
- Hypertension, heart disease, and smoking status show strong, statistically significant associations with stroke.
- BMI is also a key factor — stroke patients tend to have higher BMI values, indicating obesity as a contributing risk factor.
- Age displays a non-random increasing trend, suggesting that stroke risk rises with age.
- Gender and residence type (urban/rural) do not show significant relationships with stroke occurrence.
- Differences in BMI across work type and smoking groups further highlight lifestyle influences on stroke risk.
- Overall, Hypertension, heart disease, higher BMI, smoking habits, and older age are the most significant factors associated with an increased likelihood of stroke.

THANK YOU